Lecture 2
Solutions

Homework 2: Snapshots

Solutions to Homework 2

Run from the course folder, then work inside `practice/hw2`:

```
$ bash codeHints/L02/setup_hw2.sh
$ cd practice/hw2
$ git add README.md notes.txt
$ git commit -m "Add README and notes"
$ git add todo.txt
$ git commit -m "Add todo list"
$ echo "debug.log" > .gitignore
$ echo "secret.env" >> .gitignore
$ echo "build/" >> .gitignore
$ git add .gitignore
$ git commit -m "Add .gitignore"
$ echo "Second note." >> notes.txt
$ echo "- practice commits" >> todo.txt
$ git add notes.txt
$ git status --short
M  notes.txt
 M todo.txt
$ echo "staging area" > ../hw2_areas.txt
```

2.1. Staging only the two named files leaves `todo.txt` untracked, so it is not in the first commit. `git add .` here would have wrongly included `todo.txt`, `debug.log`, `secret.env` and `build/`. If you made a mistake before committing, `git status` shows what is staged; you will learn to unstage in Lecture 3.

2.2. Commit `todo.txt` on its own, so the history reads naturally.

2.3. Create the ignore file *before* anything adds the secrets. Check that `git status` no longer mentions the three items. (Optional: `git check-ignore -v debug.log secret.env build/output.bin` prints which rule matched each file.) `*.log` instead of `debug.log` also passes the checker.

2.4. The *first* column of `git status --short` describes the staging area, the *second* the working directory. `M  notes.txt` means "staged change"; `M todo.txt` means "modified but not staged". A commit now would include only `notes.txt`. (Do not commit: the checker expects this state.)

2.5. The **staging area**, also called the **index**. The answer file sits in `practice/`, outside `hw2`, so that it does not appear as an untracked file in the repository.

Common mistakes.

- Using `git add .` too early, which stages `secret.env` and the build files. Look at `git status` first.
- Wrong commit messages: the checker compares them exactly, including capital letters.
- Committing in task 2.4. If you did, the checker will report four commits. You can run `setup_hw2.sh --reset` and do the tasks again.
- Forgetting `-m`: an editor opens. Press `Esc`, type `:q!`, press Enter, and try again.